# ONLINE CLASSROOM CONTACT PROGRAMME

(Academic Session : 2025 - 2026)

## Enthusiast COURSE

### PHASE : ALL

### TARGET : Pre-Medical 2026

Test Type :**MAJOR** Test Pattern :**NEET (UG)**

### TEST DATE : 16-05-2026

## ANSWER KEY

| Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 2 | 2 | 4 | 3 | 1 | 1 | 3 | 2 | 3 | 4 | 4 | 3 | 4 | 3 | 2 | 1 | 2 | 3 | 2 | 2 | 2 | 4 | 2 | 4 | 1 | 2 | 4 | 4 | 1 | 1 |
| Q. | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| A. | 3 | 2 | 1 | 4 | 3 | 1 | 3 | 3 | 2 | 1 | 3 | 3 | 2 | 4 | 2 | 3 | 3 | 4 | 4 | 3 | 1 | 1 | 2 | 2 | 1 | 2 | 2 | 4 | 1 | 1 |
| Q. | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| A. | 1 | 4 | 4 | 4 | 4 | 3 | 3 | 4 | 3 | 3 | 1 | 1 | 4 | 2 | 2 | 2 | 2 | 1 | 4 | 1 | 2 | 1 | 3 | 2 | 2 | 1 | 4 | 3 | 2 | 3 |
| Q. | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 |
| A. | 2 | 3 | 3 | 3 | 1 | 4 | 3 | 2 | 3 | 3 | 3 | 1 | 3 | 1 | 4 | 2 | 2 | 1 | 2 | 3 | 1 | 1 | 3 | 4 | 1 | 1 | 4 | 2 | 3 | 3 |
| Q. | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 | 131 | 132 | 133 | 134 | 135 | 136 | 137 | 138 | 139 | 140 | 141 | 142 | 143 | 144 | 145 | 146 | 147 | 148 | 149 | 150 |
| A. | 3 | 1 | 1 | 2 | 3 | 3 | 3 | 1 | 3 | 2 | 1 | 3 | 4 | 4 | 2 | 3 | 3 | 4 | 4 | 2 | 1 | 3 | 3 | 1 | 4 | 1 | 3 | 1 | 2 | 3 |
| Q. | 151 | 152 | 153 | 154 | 155 | 156 | 157 | 158 | 159 | 160 | 161 | 162 | 163 | 164 | 165 | 166 | 167 | 168 | 169 | 170 | 171 | 172 | 173 | 174 | 175 | 176 | 177 | 178 | 179 | 180 |
| A. | 4 | 4 | 3 | 4 | 4 | 3 | 2 | 1 | 3 | 4 | 2 | 3 | 1 | 1 | 1 | 2 | 2 | 2 | 4 | 1 | 1 | 1 | 1 | 1 | 3 | 3 | 2 | 4 | 2 | 4 |

## HINT – SHEET

**1. Ans ( 2 )**

Here, $u = 144 \text{ km h}^{-1} = 144 \times \frac{5}{18} \text{ ms}^{-1} = 40 \text{ ms}^{-1}$

$v = 0,\ S = 200 \text{ m}$

As $v^2 - u^2 = 2aS$

$(0)^2 - (40)^2 = 2 \times a \times (200)$

$a = -\frac{(40)^2}{2 \times 200} = -4 \text{ ms}^{-2}$

As $v = u + at$

$\therefore \quad 0 = 40 - (4)(t)$

$t = \frac{40}{4} = 10\text{s}$

OR

$S = \left(\frac{u+V}{2}\right) t$

$200 = \left[\frac{40+0}{20}\right] t$

$t = \frac{200 \times 2}{40}$

$t = 10 \text{ sec}$

**2. Ans ( 2 )**

For triangular lamina $I = \frac{m\ell^2}{12}$

**3. Ans ( 4 )**

As the inclined plane is smooth, the sphere can never roll rather it will just slip down.

Hence, the angular momentum remains conserved about any point on a line parallel to the inclined plane and passing through the centre of the ball.

**4. Ans ( 3 )**

As $T = m\omega^2 R$

$\Rightarrow \quad 400 = 1 \times t^2 \times 1$

$\Rightarrow \quad t = 20 \text{ sec.}$

**5. Ans ( 1 )**

$g_{eff} = g - \omega^2 R$ on equator

If $g_{eff} = 0 \Rightarrow \omega^2 = \frac{g}{R}$

$\omega^2 = \frac{10}{6400 \times 10^3} = \frac{1}{64 \times 10^4} = \left(\frac{1}{800}\right)^2$

$\omega = \frac{1}{800} \text{ rad/s}$

**6. Ans ( 1 )**

As per theory

for temperature change

$Q = ms\Delta T \Rightarrow Q \propto \Delta T$

for phase change

$Q = mL \Rightarrow Q = \text{constant}$

**7. Ans ( 3 )**

As per stefan's law, $P = \sigma AT^4$

$$\frac{P_{star}}{P_{sun}} = \left(\frac{A_{star}}{A_{sun}}\right)\left(\frac{T_{star}}{T_{sun}}\right)^4$$

$$10000 = \left(\frac{r_{star}^2}{r_{sun}^2}\right)\left(\frac{1}{3}\right)^4$$

$$\Rightarrow \frac{r_{star}}{r_{sun}} = \frac{900}{1}$$

**8. Ans ( 2 )**

$$\eta = \left(1 - \frac{T_2}{T_1}\right) \times 100$$

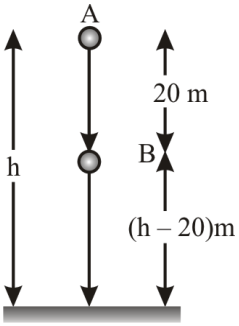
$$\Rightarrow \frac{T_2}{T_1} = 0$$

$(T_2 = 0K)$

**9. Ans ( 3 )**

For first ball $h = \frac{1}{2}gt^2$ …(i)

For second ball $(h - 20) = \frac{1}{2}g(t-1)^2$ …(ii)




subtract equation (ii) from (i)

$20 = -5 + 10t \Rightarrow \therefore t = 2.5$ sec

Hence, height $h = \frac{1}{2} \times 10 \times (2.5)^2 = 31.2$m

**10. Ans ( 4 )**

Instantaneous speed = slope of distance v/s time graph

$\because$ Slope at C is maximum

$\therefore$ speed is maximum at C

**11. Ans ( 4 )**

$$U_i = -\frac{M}{3}g\frac{L}{6} = -\frac{MgL}{18} \quad U_f = 0$$

$$W_{ext} = \;|\Delta U| = U_f - U_i = \frac{MgL}{18}$$

**12. Ans ( 3 )**

Force = Rate of change of momentum

Initial momentum $p_1 = mv\sin\theta\hat{i} + mv\cos\theta\hat{j}$

Final momentum $p_2 = -mv\sin\theta\hat{i} + mv\cos\theta\hat{j}$

$$\therefore \quad F = \frac{\Delta p}{\Delta t} = \frac{-2mv\sin\theta}{2 \times 10^{-3}}$$

Substituting m = 0.1 kg, v = 5 m/s, $\theta = 60°$

Force on the ball $F = -250\sqrt{3}$N

Negative sign indicates direction of the force.

**13. Ans ( 4 )**

Liquid with low density float over high density.

**14. Ans ( 3 )**

$$W = T(2\Delta A) = T\left[2 \times 4\pi(r_2^2 - r_1^2)\right]$$

$$W = 8\pi T(r_2^2 - r_1^2) = 8\pi T\left[\left(\frac{2}{\sqrt{\pi}}\right)^2 - \left(\frac{1}{\sqrt{\pi}}\right)^2\right]$$

$$\therefore \quad W = 8 \times \pi \times 30 \times \frac{3}{\pi} = 720 \text{ erg}$$

**15. Ans ( 2 )**

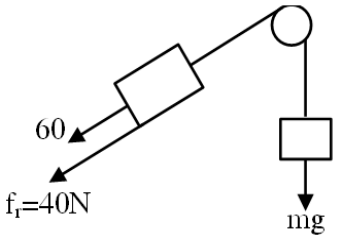
$$\because T = 2\pi\sqrt{\frac{m}{K}} \Rightarrow K \propto \frac{1}{T^2}$$

In this case $K = K_1 + K_2$ (parallel combination)

$$\frac{1}{t_0^2} = \frac{1}{t_1^2} + \frac{1}{t_2^2} \quad \Rightarrow \quad t_0^{-2} = t_1^{-2} + t_2^{-2}$$

**16. Ans ( 1 )**

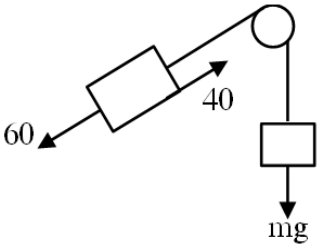
For maximum 'm'




mg = 60+40 = 100

m = 10 kg

For minimum 'm'




mg + 40 = 60

m = 2kg

**17. Ans ( 2 )**

$y_1 = a \sin(kx - \omega t)$

$y_2 = a \sin(kx + \omega t)$

According to the principle of superposition, the resultant wave is

$y = y_1 + y_2$

$= a \sin(kx - \omega t) + a \sin(kx + \omega t)$

Using trigonometric identity

$\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$

we get

$y = 2a \sin kx \cos \omega t$

**18. Ans ( 3 )**

$$\frac{a_1}{a_2} = \frac{4}{3} \Rightarrow \frac{I_1}{I_2} = \frac{a_1^2}{a_2^2} = \frac{16}{9}$$

$$\frac{I_{max}}{I_{min}} = \frac{\left(\sqrt{I_1} + \sqrt{I_2}\right)^2}{\left(\sqrt{I_1} - \sqrt{I_2}\right)^2} = \frac{49}{1}$$

**19. Ans ( 2 )**

$$m_{mean} = \frac{2.46 + 2.47 + 2.41 + 2.42 + 2.44}{5}$$

$m_{mean} = 2.44$ kg

$\Delta m_1 = 2.44 - 2.46 = -0.02$ kg

$\Delta m_2 = -0.03$ kg, $\Delta m_3 = 0.03$ kg, $\Delta m_4 = 0.02$ kg

$\Delta m_5 = 0.00$ kg

Mean error

$$= \frac{1}{5} [|-0.02| + |-0.03| + 0.03 + 0.02 + 0.00]$$

= 0.02 kg

m = 2.44 ± 0.02 kg

**20. Ans ( 2 )**

The weight of straw will be balanced by the force of surface tension $\therefore\ mg = 2Tl \Rightarrow$

$$m = \frac{2Tl}{g}$$

$$= \frac{2 \times 3 \times 10^{-2} \times 10 \times 10^{-2}}{9.8} kg = 0.6 gm$$

**21. Ans ( 2 )**

$$a = \frac{100}{10 + 10 + 5} = 4 m/s^2$$

force on B = 5 × a = 5 × 4 = 20 N

**22. Ans ( 4 )**

Before cutting string

**For Block B (mass m):**

The downward gravitational force is balanced by the upward tension (T) from the string.

T = mg

**For Block A (mass 2m):**

The downward forces are its own weight (2mg) and the tension (T) from the string pulling it down.

These are balanced by the upward spring force ($F_s$).

$F_s = 2mg + T$

Substituting T = mg:

$F_s = 2mg + mg = 3mg$

Just after cutting the string

T = 0

- Block B falls freely, so $a_B$ = g(downwards)
- Acceleration of A.

$$a_A = \frac{F_s - 2mg}{2m} = \frac{3mg - 2mg}{2m} = \frac{mg}{2m} = \frac{g}{2}$$

(upward)

**23. Ans ( 2 )**

As per wien's displacement law,

$$\lambda_m \propto \frac{1}{T} \Rightarrow \lambda m_1 T_1 = \lambda m_2 T_2$$

$$\Rightarrow T_2 = \frac{\lambda m_1 T_1}{\lambda m_2} = \frac{1.4\times10^{-6}\times1000}{2.8\times10^{-6}} = 500K$$

**24. Ans ( 4 )**

Percentage error in A

$$= 2\left(\frac{\Delta a}{a}\right) + 3\left(\frac{\Delta b}{b}\right) + \frac{\Delta c}{c} + \frac{1}{2}\left(\frac{\Delta d}{d}\right)$$

$$= 2(0.1\%) + 3(0.3\%) + 0.2\% + \frac{1}{2}(0.2\%)$$

$$= 0.2\% + 0.9\% + 0.2\% + 0.1\% = 1.4\%$$

**25. Ans ( 1 )**

for dia magnetic

$\mu_r < 1$, $\chi < 0$

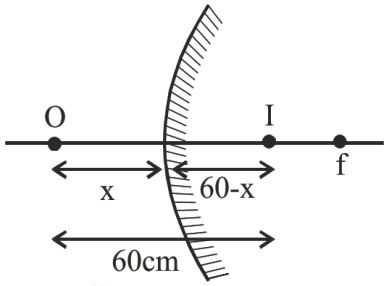
**26. Ans ( 2 )**

$$R = \frac{\sqrt{2mk}}{qB}$$

$$R' = \frac{\sqrt{2m(2k)}}{q(3B)}$$

$$\frac{R'}{R} = \frac{\sqrt{2}}{3}$$

**27. Ans ( 4 )**




$$m = -\frac{v}{u}$$

$$\frac{1}{4} = -\frac{60-x}{-x} \quad \Rightarrow x = 48cm$$

$u = 48$ cm

$v = 12$ cm

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{-48} + \frac{1}{12} = \frac{1}{f}$$

$f = 16$ cm

**28. Ans ( 4 )**

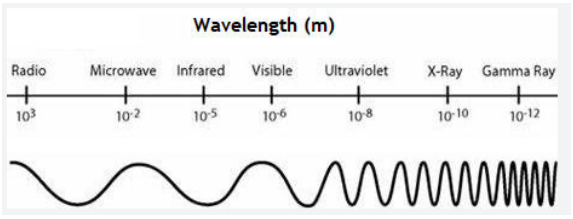
$${}_{90}X^{200} \xrightarrow{n\alpha,\, n'\beta} {}_{80}Y^{168}$$

$$n = \frac{\Delta A}{4} = \frac{200-168}{4} = \frac{32}{4} = 8$$

$${}_{90}X^{200} \xrightarrow{8\alpha} {}_{74}Z^{168} \longrightarrow {}_{80}Y^{168}$$

$n' = 80 - 74 = 6$

**29. Ans ( 1 )**




microwave oven is based on microwave.

**30. Ans ( 1 )**

$\because a\sin\theta = n\lambda$

$a\sin\theta = 2\lambda$ (for second dark)

$$\lambda = \frac{a\sin\theta}{2} = \frac{24\times10^{-7}\times\sin30^\circ}{2}$$

$= 6000$ Å

**31. Ans ( 3 )**

$$i = \frac{V}{R} = \frac{10}{2} = 5A$$

$$U = \frac{1}{2}Li^2 = \frac{1}{2}\times2\times25 = 25\ J$$

**32. Ans ( 2 )**

at resonance, $f = \frac{1}{2\pi\sqrt{LC}}$

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{80\times10^{-6}\times5}} = \frac{25}{\pi}$$

**33. Ans ( 1 )**

$\theta = (\mu_v - \mu_R)\,A = (1.68 - 1.56)\times12^\circ = 1.44^\circ$

**34. Ans ( 4 )**

In I$^{st}$ case : $H = I^2Rt \Rightarrow H \propto I^2t$ ; $t_1 = 20$ seconds, $t_2 = 18$ seconds

$$\Rightarrow \frac{H_2}{H_1} = \left(\frac{I_2}{I_1}\right)^2\frac{t_2}{t_1}$$

$$\Rightarrow \frac{H_2}{600} = \left(\frac{4}{3}\right)^2\times\frac{18}{20} \Rightarrow H_2 = 960J$$

**35. Ans ( 3 )**

As inward B decreases, so CW current in bigger loop & ACW current in smaller loop.

**36. Ans ( 1 )**

$$\int_0^A dV = -\int_0^2 Edx$$

$$V_A - V_0 = -\int_0^2 30x^2dx = -30\left[\frac{x^3}{3}\right]_0^2$$

$$= \; -30 \times \frac{8}{3}$$

$$= -80 \text{ Volt}$$

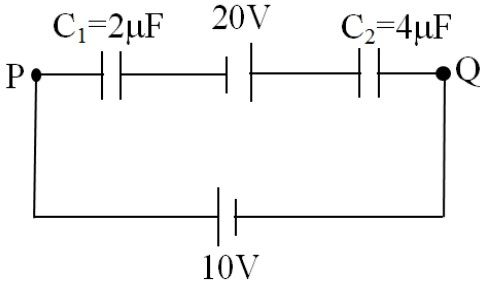
**37. Ans ( 3 )**

$$I = neAv = ne\pi r^2 v$$

$$I' = ne\pi\left(\frac{r}{2}\right)^2(2v) = \frac{I}{2}$$

**38. Ans ( 3 )**

Both battery are in series & both capacitors are also in series as per shown figure.




$$\therefore \; V_{C_1} = \frac{Q}{C_1} = \frac{C_{eff} \times V_{eff}}{C_1}$$

$$= \frac{\frac{C_1C_2}{C_1+C_2} \times V\,eff}{C_1} = \frac{C_2}{C_1 + C_2} \times V_{eff}$$

$$= \frac{4}{2+4} \times 30 = 20 \text{ V}$$

**39. Ans ( 2 )**

$$K_{max}(eV) = 12400\left[\frac{1}{\lambda(Å)} - \frac{1}{\lambda_0(Å)}\right]$$

$$= 12400\left[\frac{1}{1000} - \frac{1}{2000}\right] = 6.2 \text{ eV}$$

**40. Ans ( 1 )**

$$\lambda = \frac{h}{\sqrt{3mKT}}$$

$$\lambda \propto \frac{1}{T}$$

$$\frac{\lambda_2}{\lambda_1} = \sqrt{\frac{T_1}{T_2}}$$

$$\lambda_2 = \lambda\sqrt{\frac{300}{1200}}$$

$$= \frac{\lambda}{2}$$

**41. Ans ( 3 )**

$$n\lambda_1 = (n + 1)\lambda_2$$

$$7500\, n = 6000\,(n + 1)$$

$$5n = 4n + 4$$

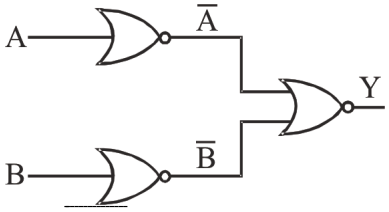
$$n = 4$$

**42. Ans ( 3 )**

$$R \propto A^{1/3} \Rightarrow \frac{R_1}{R_2} = \left(\frac{16}{128}\right)^{1/3} \Rightarrow \frac{3 \times 10^{-15}}{R_2} = \frac{1}{2}$$

$$R_2 = 6 \times 10^{-15} \text{ m}$$

**43. Ans ( 2 )**




$$Y = \bar{A} + \bar{B}$$

$$Y = \overline{\bar{A} + \bar{B}} = A.B \quad \text{(AND)}$$

**44. Ans ( 4 )**

$$\because \; \sin\theta_c = \frac{\mu_r}{\mu_d} \text{ and } \mu = \frac{c}{v}$$

$$\text{So, } \sin\theta_c = \frac{v_d}{v_r}$$

$$\sin\theta_C = \frac{V_D}{V_r} = \frac{1.8}{2.4} = \frac{3}{4}$$

$$\theta_C = \sin^{-1}\frac{3}{4}$$

**45. Ans ( 2 )**

$$I_{Load} = \frac{6}{1 \times 10^3} = 6\text{mA}$$

$$I_{100\Omega} = \frac{9-6}{100} = 30 \text{ mA}$$

$$I_Z = I_{100} - I_L = 30 - 6 = 24 \text{ mA}$$

**46. Ans ( 3 )**

$$\alpha = \sqrt{\frac{K_a}{C}}$$

$$\Rightarrow \alpha = \sqrt{\frac{1.4 \times 10^{-5}}{(0.1/2)}}$$

$$\Rightarrow \alpha = 1.67 \times 10^{-2}$$

**47. Ans ( 3 )**

Formation of 1 mole of compound from element at elementry form

$$C(\text{graphite, s}) + 2H_2(g) \longrightarrow CH_4(g)$$

**48. Ans ( 4 )**

Charge required to deposite 1 gm equiv. of any substance is 1F

**49. Ans ( 4 )**

$$\text{V.P.} \propto \frac{1}{\text{b. p.}}$$

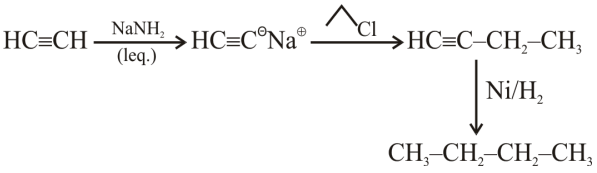
Solution III has lowest

Value of V.P. so highest b.p.

**50. Ans ( 3 )**

Geometrical isomers are diastereomers also

**51. Ans ( 1 )**



**52. Ans ( 1 )**

As per trans effect.

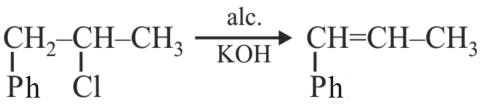
**53. Ans ( 2 )**

$H_2SO_5$ oxidation state of sulphur +6.

**54. Ans ( 2 )**

$$E^\circ_{cell} = E^\circ_{op} + E^\circ_{rp}$$

**55. Ans ( 1 )**



**56. Ans ( 2 )**

B.E. $\propto$ Stability

**57. Ans ( 2 )**

Stability in aqueous

$Cr^{3+} > Mn^{3+}$

$CrO_4^{2-} \xrightarrow{H^+} Cr_2O_7^{2-}$

yellow orange

Brass = Cu + Zn

Bronze = Cu + Sn

**58. Ans ( 4 )**

(A) $ns^2\ np^1$ – Group 13

(B) $ns^2\ np^2$ – Group 14

(C) $ns^2\ np^3$ – Group 15

(D) $ns^2\ np^4$ – Group 16

**59. Ans ( 1 )**

**Question Explanation:** Identify which given compounds/species are *chemically possible (can exist).*

**Concept:** Octet rule, expansion of octet, stability of ions.

**Solution:**

- **$H_2S$** ⟶ Exists (bent molecule).
- **$OF_6$** ⟶ Does *not* exist (O cannot expand octet beyond $8e^-$).
- **$BF_4^-$** ⟶ Exists (stable tetrahedral ion).
- **$PH_5$** ⟶ Does *not* exist (P–H bonds cannot stabilize hypervalent form).
- **$XeF_5$** ⟶ Does *not* exist ($XeF_4$ and $XeF_6$ exist, not $XeF_5$).
- **$CCl_6^{2-}$** ⟶ Does *not* exist (carbon can't form six bonds).
- **$PCl_5$** ⟶ Exists (stable in gaseous state, trigonal bipyramidal).

**Existing species:** $H_2S$, $BF_4^-$, $PCl_5$ ⟶ **3 compounds**

**Final Answer: Option (1)**

**60. Ans ( 1 )**

(A) $[Co(NH_3)_5(NO_2)]Cl_2$ Linkage

$NO_2^-$ can be converted into $ONO^-$

(B) $[Co(NH_3)_5SO_4]Br$ Ionisation

$[Co(NH_3)_5Br]\ SO_4$

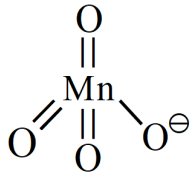
(C) $[Co(NH_3)_6]\ [Cr(CN)_6]$ Coordination

(D) $[Co(H_2O)_6]Cl_3$ Solvate

**61. Ans ( 1 )**

$KMnO_4$

$K^+\ MnO_4^-$



$Mn^{+7} \longrightarrow d^\circ$ configuration

diamagnetic

**62.** **Ans ( 4 )**

$$\overset{+4\ -1}{PbI_4} \xrightarrow{\text{Redox}} \overset{+2}{PbI_2} + I_2$$

(O.A.)(R.A.)

**63.** **Ans ( 4 )**

Glucose gives positive test with fehling solution.

**64.** **Ans ( 4 )**

$$CH_3COOH + PCl_5 \longrightarrow CH_3-\underset{\underset{O}{\|}}{C}-Cl \xrightarrow[\text{dry ether}]{CH_3MgBr}$$

$$CH_3-\underset{\underset{O}{\|}}{C}-CH_3 \xrightarrow[(2)\ H_2O]{(1)\ CH_3MgBr} CH_3-\underset{\underset{OH}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_3$$

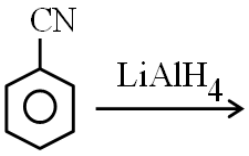
Product

**65.** **Ans ( 4 )**

Reagent : $NH_2-NH-\overset{\overset{O}{\|}}{C}-NH_2$

Semicarbazide

**66.** **Ans ( 3 )**



**67.** **Ans ( 3 )**

Acidic strength $\propto \frac{EWG}{ERG}$

d > c > a > b

**68.** **Ans ( 4 )**

$$(CH_3)_2CH_2 \xrightarrow[\text{Oxidation}]{KMnO_4} (CH_3)_2CH-OH$$

Only 3° hydrogen to convert OH.

**69.** **Ans ( 3 )**

| (A) | (B) | (C) | (D) | (E) |
|---|---|---|---|---|
| (i) | (iv) | (iii) | (v) | (ii) |

**70.** **Ans ( 3 )**

(i) & (iii) only

(i) S-orbital - spherical

(ii) Non directional

**71.** **Ans ( 1 )**

$$K = \sqrt{\frac{K_1}{K_2}}$$

**72.** **Ans ( 1 )**

**Generated by Allie**

**Problem Statement:** The question asks to match terms related to collision theory in chemistry. It requires linking definitions or descriptions in Column-I with the correct terms or concepts in Column-II that describe aspects of molecular collisions and reaction kinetics.

**Underlying Concept:** The core concept is collision theory in chemical kinetics. It explains how chemical reactions occur through collisions between molecules. Key ideas include collision frequency, activation energy, effective collisions, and deviations for complex molecules.

**Relevant Formulas:** The relevant formula is the rate expression from collision theory:

Where: - $Z_{AB}$ = collision frequency (number of collisions per second per unit volume) - $E_a$ = activation energy - R = gas constant - T = temperature - $e^{-E_a/RT}$ = fraction of molecules with energy equal to or greater than $E_a$ (Boltzmann factor)

Effective collisions lead to reaction, depending on sufficient kinetic energy and proper orientation.

**Tips and Tricks:** Remember the order: number of collisions (frequency), fraction energetic enough (Boltzmann factor), adjustments for complex molecules, then effective collisions that cause reactions.

**Common Mistakes:** Confusing collision frequency with effective collisions. Not identifying the Boltzmann factor as the fraction of molecules with enough energy. Overlooking deviations caused by complex molecules.

**Why Other Options Are Incorrect?:** Options mixing collision frequency with effective collisions or mismatching complex molecules lead to misunderstanding of reaction kinetics and collision theory.

**73.** **Ans ( 4 )**

**Explain question:**

To select the correct match.

**Concept:** Azeotropic Mixture

**Solution:**

(A) 68% $HNO_3$+ 32% $H_2O$ ⟶ This mixture forms a maximum boiling azeotrope because it shows negative deviation from Raoult's law. (A ⟶ III).

(B) Solution having large positive deviation: it form minimum boiling azeotropes (B ⟶ I).

(C) n-heptane and n-hexane : It forms ideal solution (C ⟶ II).

(D) Two solutions having the same osmotic pressure at a given temperature: are known as isotonic solution (D ⟶ IV).

**Final answer:** Option (4)

**74.** **Ans ( 2 )**

$\Lambda_m^o(NH_4OH) = \Lambda_m^o(NH_4Cl) + \Lambda_m^o(NaOH) - \Lambda_m^o(NaCl)$

$\alpha = \frac{\Lambda_m}{\Lambda_m^o}$

**75.** **Ans ( 2 )**

(A) $[\underline{C}r(NH_3)_4Cl_2]^+$ = +3

(B) $\underline{N}H_4Cl$ = –3

(C) $H_2\underline{S}_2O_8$ = +6

(D) $\underline{Os}O_4$ = +8

**76.** **Ans ( 2 )**

(A) Hs ⟶ 108

(B) Rg ⟶ 111

(C) Mc ⟶ 115

(D) Og ⟶ 118

**77.** **Ans ( 2 )**

Regular geometry have zero dipole moment.

$CF_4$, $BF_3$, $CO_2$, $PCl_5$

**78.** **Ans ( 1 )**

Both Assertion and Reason are true and the Reason is the correct explanation of Assertion

**79.** **Ans ( 4 )**

$Mn^{+2} \longrightarrow 3d^5 \longrightarrow$ stable

**80.** **Ans ( 1 )**

$Mn^{+2} \longrightarrow d^5$

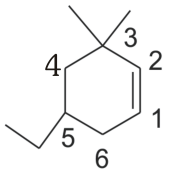
| | | | | |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 |

**81.** **Ans ( 2 )**

Colour of $BaCrO_4$ is yellow

$H_2S$ is a weak acid it does not produce free $S^{2-}$ ion

**82.** **Ans ( 1 )**




5-ethyl-3,3-dimethyl cyclohexane

**83.** **Ans ( 3 )**

Resonating structure

**84.** **Ans ( 2 )**

$C_2H_5–NH–C_2H_5$ is a secondary amine, therefore, it will not give carbylamine test.

**85.** **Ans ( 2 )**

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$V_2 = \frac{P_1V_1T_2}{P_2T_1} = \frac{700 \times 40 \times 273}{760 \times 300}$

$V_2 = 33.52$

22400mL of $N_2$ consis 28 g then 33.52 mL g

$N_2$ confain

$\frac{33.52 \times 28}{22400}$

$N\% = \frac{0.0419}{0.25} \times 100$

=16.76

**86.** **Ans ( 1 )**

$SN^2$ mechanism $CH_3 > 1° > 2° > 3°$

87. **Ans ( 4 )**

$AgCl \rightleftharpoons Ag^+ + Cl^-$

$K_{sp} = (Ag^+)\,(Cl^-)$

$(Ag^+) = \left[\frac{K_{sp}}{(Cl^-)}\right] = \frac{1.6 \times 10^{-10}}{0.2} = 8 \times 10^{-10}\ M$

88. **Ans ( 3 )**

For zero order reaction

$[A]_t = [A]_0 - Kt$

89. **Ans ( 2 )**

$\frac{P_A^o - P_S}{P_S} = \frac{n_B}{n_A}$

$\Delta T_f = K_f.m$

90. **Ans ( 3 )**

(A) In titration of $KMnO_4$ and oxalic acid, $KMnO_4$ act as self indicator ⟶ Correct

(C) A solution of known concentration is called a standard solution. ⟶ Correct

(D) A lyophilic sol is more stable than lyophobic sol. ⟶ Correct

91. **Ans ( 2 )**

**Question explanation :** Identify the correct statement about the epithelial tissue shown in the diagram.

**Concept:** compound epithelium

**Solution:**

(1) It is composed of tall and single layered cells ⟶ Incorrect. That describes epithelium, is stratified (compound)in which
- cells are multilayered not single layer.

(2) It has limited role in secretion and absorption ⟶ Correct. Stratified squamous epithelium mainly provides protection, not
- secretion/absorption.

(3) It is found in nephron ⟶ Incorrect. Nephron has cuboidal and squamous
- epithelium, but not stratified squamous.

(4) Their main function is to form diffusion boundaries ⟶ Incorrect. That is the role of simple squamous epithelium (alveoli,
- capillaries).

Correct statement = Option 2

**Final Answer: Option 2**

92. **Ans ( 3 )**

**Question explanation :** Identify the abdominal segments in which testes are present in male cockroach

**Concept :** Cockroach

**Solution :** The correct answer is option 3

In male cockroach, a pair of testes are situated in the 4th to 6th abdominal segments. They form the primary
- reproductive organs and produce sperms.

**Final answer : Option (3)**

93. **Ans ( 3 )**

**Question explanation :** Ask which respiratory organs are involved in respiration of frog on land.

**Concept :** Respiration in frog

**Solution :** The correct answer is option (3)

Adult frog shows three types of respiration on land:
- Cutaneous respiration through moist skin
- Buccopharyngeal respiration through lining
- of buccal cavity
- Pulmonary respiration through lungs

The moist skin helps in diffusion of gases, while lungs help in efficient breathing on land. Gills are present only in tadpole stage, not in adult frog.

**Final answer : Option (3)**

94. **Ans ( 3 )**

**Question explanation :** Identify the common feature present in both Periplaneta and Pheretima.

**Concept :** Animal Kingdom

**Solution :** The correct answer is option (3) Only D.

Both Periplaneta (cockroach) and Pheretima (earthworm) possess a double ventral nerve cord.

(A) Open blood vascular system ⟶ Present in cockroach but earthworm has closed
- circulatory system.

(B) Internal fertilisation ⟶ Cockroach shows internal fertilisation, while earthworm
- shows external fertilisation in cocoon.

(C) Dioecious ⟶ Cockroach is dioecious, but earthworm is monoecious
- (hermaphrodite).

(D) Double ventral nerve cord ⟶ Present in
- both organisms.

**Final answer :** Option (3)

95. Ans ( 1 )

**Question explanation :** Identify the incorrect statement related to phylum Platyhelminthes.

**Concept :** Animal Kingdom - Platyhelminthes

**Solution :** The correct answer is option (1) All are endoparasite.

All platyhelminthes are not endoparasites because some members like Planaria are free-living organisms.

- They are bilaterally symmetrical and triploblastic.
- Parasitic forms possess hooks and suckers for attachment to host body.
- Their body is dorso-ventrally flattened, therefore they are called flatworms.

Hence, statement (1) is incorrect.

**Final answer : Option (1)**

96. Ans ( 4 )

**Question explanation** : Identify the type of joint present between carpal and metacarpal bones of thumb.

**Concept** : Joints

**Solution** : The correct answer is option (4) Saddle.

The joint between the carpal and metacarpal of thumb is a saddle joint. It allows movement in different directions, giving the thumb great flexibility and opposition movement.

- Ball and socket joint is present in shoulder and hip.
- Hinge joint is present in elbow and knee.
- Pivot joint is present between atlas and axis vertebrae.

**Final answer :** Option (4)

97. Ans ( 3 )

**Question explanation :** Identify the effect of vasoconstriction on blood pressure.

**Concept :** Circulation

**Solution :** The correct answer is option (3) Increase in blood pressure.

- Vasoconstriction means narrowing of blood vessels due to contraction of smooth muscles in vessel walls. This reduces the diameter of blood vessels and increases resistance to blood flow, leading to an increase in blood pressure.

**Final answer** : Option (3)

98. Ans ( 2 )

**Question explanation :** Ask how many given statements about circulation and portal vein are wrong.

**Concept:** Circulation

**Solution:** The correct answer is option :2

- (A) In amphibia and reptiles incomplete double circulation present ⟶ Correct. Amphibians and reptiles have three-chambered hearts (except crocodile), so circulation is incomplete.
- (B) In arthropods and molluscs open circulatory system is present ⟶ Correct. Arthropods and most molluscs have open circulation (except cephalopods).
- (C) In joint diastole stage semilunar valves are closed ⟶ Correct. During joint diastole, both atria and ventricles relax, semilunar valves remain closed.
- (D) The hepatic portal vein carries blood from intestine to the kidney ⟶ Incorrect. Hepatic portal vein carries blood from intestine to liver, not kidney.

Only Statement D is wrong.

**Final Answer:** Option 2

99. Ans ( 3 )

**Question explanation** : Identify the components that form Functional Residual Capacity (FRC).

**Concept** : Respiratory volumes

**Solution** : The correct answer is option 3.

Functional Residual Capacity (FRC) is the volume of air remaining in lungs after a normal expiration. It is calculated by adding:

- ERV (Expiratory Reserve Volume)
- RV (Residual Volume)

**Final answer :** Option (3)

100. Ans ( 3 )

**Question explanation :** Identify the respiratory centre present in pons of brain.

**Concept :** Respiratory centres

**Solution :** The correct answer is option (3)

Pneumotaxic centre.

The pneumotaxic centre is present in the pons region of brain and helps in regulating the respiratory rhythm by controlling the duration of inspiration.

- Dorsal and ventral respiratory centres are present in medulla oblongata.
- Chemosensitive centre is also associated with medulla and responds to changes in $CO_2$ and $H^+$ concentration.

**Final answer :** Option (3)

101. Ans ( 3 )

**Question explanation :** Identify whether the given statements about synapse and resting potential are correct or incorrect.

**Concept** : Neural

**Solution :** The correct answer is option 3

- Statement-I: In chemical synapses, neurons are separated by a small gap called synaptic cleft, through which neurotransmitters diffuse. Hence, the statement is correct.
- Statement-II: The electrical potential difference across the resting plasma membrane of a neuron is known as resting potential. Hence, this statement is also correct.

Therefore, both statements are correct.

**Final answer** : Option (3)

102. Ans ( 1 )

**Question explanation :** Identify the permeability of axonal membrane during resting state of neuron.

**Concept** : Neural

**Solution** : The correct answer is option 1

- During resting state, the axonal membrane allows more movement of $K^+$ ions to the outside, while it is almost impermeable to $Na^+$ ions. This unequal permeability helps maintain the resting potential of neuron.

**Final answer :** Option (1)

**103. Ans ( 3 )**

**Question explanation :** Identify whether the assertion and reason related to urine analysis and glycosuria are correct.

**Concept :** Excretion

**Solution :** The correct answer is option 3

**Assertion:** Analysis of urine helps in diagnosis of many metabolic disorders ⟶ Correct, because abnormal substances in urine can indicate diseases.

**Reason:** Glycosuria indicates diabetes insipidus ⟶ Incorrect, because glycosuria (presence of glucose in urine) is mainly associated with diabetes mellitus, not diabetes insipidus.

Diabetes insipidus is related to deficiency of ADH and causes excessive dilute urine without glucose.

**Final answer :** Option (3)

**104. Ans ( 1 )**

**Question explanation :** Identify the correct categorisation of animals on the basis of nitrogenous waste excretion.

**Concept :** Excretion

**Solution :** The correct answer is option 1

Animals are classified according to the main nitrogenous waste they excrete:

Ammonotelic ⟶ Excrete ammonia
- Example: Aquatic amphibians

Ureotelic ⟶ Excrete urea
- Example: Frog, Humans

Uricotelic ⟶ Excrete uric acid
- Example: Pigeon, Lizards, Cockroach

Ammonia requires large amount of water for removal, while uric acid is least toxic and conserves water.

**Final answer :** Option (1)

**105. Ans ( 4 )**

**Question explanation** : Identify the body part where ball and socket joint is present.

**Concept :** Joints

**Solution :** The correct answer is option 4

Ball and socket joint allows movement in all directions and rotation. It is present in the shoulder and hip joints.

- Wrist has gliding joint.
- Fingers have hinge joints.
- Neck has pivot joint.

**Final answer** : Option (4)

**106. Ans ( 2 )**

NCERT Page No. # 225 (E/H)

**107. Ans ( 2 )**

**Question explanation** : Identify the correct matching between hormones and their functions.

**Concept** : Hormones

**Solution :** The correct answer is option (2).

Correct matching is:

A. Growth hormone ⟶ iii. Diabetogenic hormone

Growth hormone increases blood glucose level, therefore called diabetogenic
- hormone.

B. Luteinizing hormone ⟶ i. Ovulation

LH surge induces ovulation in females.
- C. Prolactin hormone ⟶ iv. Milk formation

Prolactin stimulates milk production in
- mammary glands.

D. FSH ⟶ ii. Stimulate Sertoli cells

FSH acts on Sertoli cells and helps in
- spermatogenesis.

**Final answer :** Option (2)

108. **Ans ( 1 )**

**Question explanation :** Identify the hormone that regulates skin pigmentation by acting on melanocytes.

**Concept :** Hormones

**Solution** : The correct answer is option 1.

Melanocyte stimulating hormone (MSH) acts on melanocytes, the melanin-containing cells of skin, and regulates the synthesis and distribution of melanin pigment. This helps control skin pigmentation.

- ACTH mainly stimulates adrenal cortex.
- Growth hormone promotes body growth.
- Prolactin stimulates milk production.

**Final answer** : Option (1)

109. **Ans ( 2 )**

NCERT Page # 51

110. **Ans ( 3 )**

**Question explanation :** Identify the theory which states that spores of life came from outer space to Earth.

**Concept** : Evolution

**Solution** : The correct answer is option 3.

According to the Panspermia theory, life existed in the form of spores or seeds in space and reached Earth from other planets or celestial bodies. Early Greek thinkers supported this idea.

- Oparin-Haldane theory explains chemical evolution on Earth.
- Big Bang theory explains origin of universe.
- Abiogenesis states life originated from non-living matter on Earth.

**Final answer :** Option (3)

111. **Ans ( 1 )**

NCERT XII, Page # 130

112. **Ans ( 1 )**

**Question explanation :** Identify the hormone responsible for milk ejection from mammary glands in females.

**Concept** : Hormones

**Solution :** The correct answer is option 1.

Oxytocin stimulates contraction of smooth muscles around mammary glands, causing milk ejection or milk let-down reflex.

- Prolactin helps in milk formation, not milk ejection.
- Vasopressin regulates water balance in body.

**Final answer :** Option (1)

113. **Ans ( 3 )**

**Question explanation** : Identify the hormone that maintains functions of male accessory ducts and glands.

**Concept** : Reproduction

**Solution :** The correct answer is option 3.

Androgens, mainly testosterone secreted by Leydig cells, maintain the structure and function of male accessory ducts and glands such as seminal vesicles, prostate gland and vas deferens. They also regulate male secondary sexual characters.

- FSH stimulates spermatogenesis.
- LH/ICSH stimulates Leydig cells to secrete testosterone.

**Final answer :** Option (3)

114. **Ans ( 4 )**

NCERT-XII, Pg # 48

115. **Ans ( 1 )**

NCERT (XIIth) (Eng), Para-2, Pg. # 54

NCERT (XIIth) (Hindi), Para-1, Pg. # 59

116. **Ans ( 1 )**

**Question explanation :** Identify whether the assertion and reason related to passive immunisation are correct.

**Concept :** Immunity

**Solution :** The correct answer is option 1.

Assertion: Antivenom and antiserum are examples of passive immunisation ⟶ Correct, because ready-made antibodies are directly given to the patient.

Reason: Antivenom contains preformed antibodies against snake venom ⟶ Correct. These antibodies provide immediate protection without waiting for the body to produce its own antibodies.

Thus, the reason correctly explains the assertion.

**Final answer** : Option (1)

117. **Ans ( 4 )**

**Question explanation :** Identify the correct matching between reproductive processes and their sites.

**Concept :** Reproduction

**Solution** : The correct answer is option 4.

Correct matching is:

(A) Fertilization ⟶ (iii) Ampulla

Fertilization usually occurs at ampullary-isthmic junction of fallopian tube.

(B) Implantation ⟶ (i) Fundus of uterus

Blastocyst gets implanted in the uterine wall, generally near the fundus.

(C) Functional maturation of sperms ⟶ (iv) Epididymis

Sperms gain motility and functional maturity in epididymis.

(D) Oogenesis ⟶ (ii) Ovary

Formation of ova takes place in ovary.

**Final answer :** Option (4)

118. **Ans ( 2 )**

**Question explanation :** Identify whether the statements related to malignant tumours are correct or incorrect.

**Concept :** Cancer

**Solution** : The correct answer is option 2.

Statement-I: Malignant tumours exhibit metastasis ⟶ Correct. Malignant cancer cells spread from one part of body to another through blood or lymph.

Statement-II: Contact inhibition is the property of malignant tumours ⟶ Incorrect. Malignant tumour cells lose contact inhibition and continue uncontrolled division.

Normal cells show contact inhibition, whereas cancer cells do not.

**Final answer :** Option (2)

119. **Ans ( 3 )**

**Question explanation :** Identify the hormones secreted only during pregnancy in women.

**Concept :** Hormones

**Solution :** The correct answer is option 3.

Relaxin is secreted during pregnancy to relax pelvic ligaments.

hCG (human chorionic gonadotropin) is produced by placenta during pregnancy.

hPL (human placental lactogen) is also secreted by placenta only during pregnancy.

Oxytocin and estrogen are present even in non-pregnant women, therefore they are not pregnancy-specific hormones.

**Final answer :** Option (3)

120. Ans ( 3 )

**Question explanation** : Identify the component that is part of innate immunity.

**Concept** : Immunity

**Solution :** The correct answer is option 3.

Natural killer (NK) cells are a part of innate immunity and provide non-specific defence against virus-infected and tumour cells.

- B-lymphocytes and T-lymphocytes are components of acquired immunity.
- Antibodies are produced during specific immune response.

**Final answer** : Option (3)

121. Ans ( 3 )

**Question Explanation:** The question asks to correctly match trophic levels with their appropriate examples.

**Concept:** – Trophic Levels **Explanation:**

- Primary producers are autotrophs like grass.
- Primary consumers are herbivores like grasshopper.
- Secondary consumers are carnivores feeding on herbivores.
- Tertiary consumers are top carnivores like wolf/man.

**Final Answer:** (3) a-1-N, b-2-K, c-3-L, d-4-M

122. Ans ( 1 )

**Question Explanation:** The question asks to identify the statement that is not true regarding an ecosystem.

**Concept:** – Structure and Function **Explanation:**

- Ecosystem requires a continuous input of energy (mainly solar energy).
- Energy flow is unidirectional, not cyclic.
- Nutrients (materials) are recycled between biotic and abiotic components.
- Ecosystems are generally self-sustaining systems.

The incorrect statement is (1), because energy does not cycle in an ecosystem; only nutrients cycle.

**Final Answer:** (1) Cycling of energy occurs

123. Ans ( 1 )

NCERT XII Pg # 237

124. Ans ( 2 )

**Question Explanation:** The question asks to identify the region that is not recognized as a biodiversity hotspot in India.

**Concept:** – Biodiversity Hotspots

**Explanation:**

- Biodiversity hotspots are regions with very high species richness and high endemism,
- but also high threat levels.
- India includes parts of the following
- hotspots:
  - Himalaya
  - Indo-Burma
  - Western Ghats–Sri Lanka (Indian
  - part: Western Ghats)

The Eastern Ghats are not listed among global biodiversity hotspots.

**Final Answer:** (2) Eastern ghat

125. **Ans ( 3 )**

**Question Explanation:**

The question asks about the estimated number of mango varieties present in India.

**Concept:** Biodiversity in India

**Explanation:**

India is estimated to have around 1000

- varieties of mango.

These include both wild and cultivated types

- developed through selection and cultivation.

**Final Answer:** (3) 1000

126. **Ans ( 3 )**

**Question Explanation:** The question asks how many given statements about ecological interactions are correct.

**Concept:** – Interactions in Ecosystem **Explanation:**

- Predation helps regulate prey populations.

A significant proportion of insects are plant-eating (phytophagous). About one-fourth of

- insects feed on plants.

Parasitism involves close association; commensalism may or may not involve very

- close physical dependency.
- Endoparasites live inside the host body.

**Final Answer:** (3) 3

127. **Ans ( 3 )**

NCERT-XI, Pg. # 126

128. **Ans ( 1 )**

**Question Explanation :**

Question asking about the ribosomes

**Concept :**

Question is based on ribosomes

**Solution :**

The eukaryotic ribosomes are 80S Each ribosome has two subunits, larger and smaller subunits .The two subunits of 80S ribosomes are 60S and 40S.

**Final Answer :**

Option (1)

129. **Ans ( 3 )**

**Explaining the Question :-** The question asks about the specific phase in the cell cycle during which **DNA replication** and **centriole duplication** take place.

**Concept:-** This question is based on **Cell cycle**

**Solution :-** DNA replication and centriole duplication occur during the **S phase** of the cell cycle.

**Final Answer :-** (3)

130. **Ans ( 2 )**

NCERT XI Pg # 140

131. **Ans ( 1 )**

**Question Explanation:** The question asks to identify the statement that is NOT a significance of meiosis.

**Concept:** Cell Cycle – Meiosis Significance

**Explanation:**

It maintains constant number of chromosomes in different cells of an organism -False

This is incorrect because meiosis reduces chromosome number, and constancy in somatic cells is maintained by mitosis, not meiosis.

**Final Answer:** (1) It maintains constant number of chromosomes in different cells of an organism

132. **Ans ( 3 )**

**Question Explanation:**

The question asks about the term used for the rate of formation of new organic matter by consumers.

**Concept:** – Productivity **Explanation:**

Primary productivity refers to production by

- producers (plants).

Secondary productivity refers to production

- by consumers (heterotrophs).

Gross and net primary productivity are

- related only to producers.

**Final Answer:** (3) Secondary productivity

133. **Ans ( 4 )**

**Question Explanation:** The question asks to identify the correct mathematical expression for the species–area relationship.

**Concept:** – Species-Area Relationship

**Explanation:**

The species–area relationship describes how the number of species increases with the area

- sampled.

It is given by the equation:

- $S = C\,A^Z$

Taking logarithm on both sides:

- $\log S = \log C + Z \log A$

**Final Answer:** (4) $\log S = \log C + Z \log A$

134. **Ans ( 4 )**

NCERT Page No. # 126, 127

135. **Ans ( 2 )**

**Question Explanation:** The question asks which floral feature is represented in both floral formula and floral diagram.

**Concept:** – Floral Formula & Diagram

**Explanation:**

Floral formula represents number, symmetry, fusion (adhesion), and

- arrangement of floral parts symbolically.

Floral diagram gives a visual arrangement of

- floral whorls.

Both together represent features like fusion/adhesion and arrangement of floral parts, but not internal ovary details like

- locules or ovule arrangement.

Adhesion between parts of floral whorls Shown in both (e.g., A, G with fusion symbols in formula and joined whorls in diagram)

**Final Answer:** (2) Adhesion between parts of floral whorls

136. **Ans ( 3 )**

**Question Explanation:**

The question asks to identify the flower in which the ovary is inferior.

**Concept:** – Position of Ovary **Explanation:**

Inferior ovary: ovary is below the attachment of other floral parts (epigynous

- flower).

Superior ovary: ovary is above the attachment of other floral parts (hypogynous

- or perigynous flowers).

Guava - Ovary is inferior (epigynous flower)

**Final Answer:** (3) Guava

137. **Ans ( 3 )**

**Explain Question :** The question asks you to identify which of the listed processes is not a part of sexual reproduction.

**Concept :** This question is based on gymnosperm, general characters and life cycle.

**Explanation :** Formation of new organism from unfertilized female gamete - Not sexual reproduction

This describes parthenogenesis.

**Final answer :** 3

138. **Ans ( 4 )**

NCERT XII Pg # 23, 25, 36, 38

139. **Ans ( 4 )**

**Question Explanation:**

The question asks to identify which statements about $C_3$ plants and RuBisCO are correct.

**Concept:** – RuBisCO and $C_3$ pathway

**Explanation:**

(A) If $O_2$ binds to RuBisCO then $CO_2$ fixation decreases - True

($O_2$ competes with $CO_2$ $\rightarrow$ photorespiration

- reduces $CO_2$ fixation)

(E) RuBisCO has much greater affinity for

- $CO_2$ - True

**Final Answer:** (4) A and E

**140. Ans ( 2 )**

**Question Explanation**: Question asks about the kingdom whose organism boundaries are not well defined.

**Concept:** Protista

**Solution**: Kingdom Protista contains unicellular eukaryotic organisms. Their boundaries are unclear because they show mixed characteristics.

**Final Answer**: Option (2) Protista

**141. Ans ( 1 )**

NCERT XI Pg # 38

**142. Ans ( 3 )**

**Question Explanation:**

The question asks to identify the incorrect statement among the given options.

**Concept:** Plant Kingdom – Classification

**Explanation:**

- Brown algae show all types of sexual reproduction: isogamy, anisogamy, oogamy.
- Liverworts (Bryophytes) prefer moist and shady habitats.
- Gymnosperms have naked ovules (not enclosed by ovary).
- In pteridophytes, the main plant body is the sporophyte generation.

**Final Answer:** (3) In gymnosperms ovules are enclosed by ovary wall

**143. Ans ( 3 )**

**Question Explanation:** The question asks how many of the given statements about embryology and fruits are correct.

**Concept:** Sexual Reproduction in Flowering Plants

**Explanation:**

- Epiblast is present in grass embryo (monocots like maize/wheat).
- Endosperm is formed from the primary endosperm nucleus (3n) after double fertilization.
- Albuminous seeds retain endosperm; pea and groundnut are non-albuminous.
- False fruits develop from parts other than ovary (e.g., apple, strawberry).
- Viable seeds of Phoenix dactylifera (date palm) have been reported even after long dormancy (around 2000 years).

Final Count:

Correct statements = A, B, D, E = 4 statements

**Final Answer:** (3) Four

**144. Ans ( 1 )**

NCERT XI Pg # 222 Fig. 13.10

**145. Ans ( 4 )**

**Question Explanation:** The question asks to identify the correct substrate of the Krebs cycle and the number of oxidation reactions and $CO_2$ molecules released during aerobic respiration.

**Concept:** – Krebs Cycle / TCA Cycle **Explanation:**

- The Krebs cycle starts with Acetyl Co-A (2C compound) combining with oxaloacetate.
- During one turn of the cycle:
  - 2 $CO_2$ molecules are released
  - 4 oxidation reactions occur ($NAD^+$ and FAD reduced)
  - One cycle runs twice per glucose molecule (but values given are per acetyl Co-A cycle)

**Final Answer:** (4) Acetyl Co-A, 4 oxidation, 2 $CO_2$ release

146. **Ans ( 1 )**

**Question Explanation:** The question asks to identify the plant growth hormone responsible for initiating rooting in stem cuttings.

**Concept:** Plant Growth Hormones – Auxins

**Explanation:**

Plant hormones regulate growth and development. Different hormones have specific roles such as cell division, elongation, and differentiation.

Auxins are the primary hormones responsible for:

- Cell elongation
- Apical dominance
- Formation of adventitious roots (rooting in stem cuttings)

**Final Answer:** (1) Auxins

147. **Ans ( 3 )**

**Question Explanation:**

The question asks to identify the incorrect match between fungal class, example, and their characteristics.

**Concept:** Biological Classification – Fungi

**Explanation:**

Fungi are classified into major groups based on morphology, reproduction, and spore formation:

- Phycomycetes ⟶ mostly aquatic/terrestrial lower fungi (e.g., Albugo, Rhizopus)
- Ascomycetes ⟶ sac fungi (e.g., Penicillium, yeast)
- Basidiomycetes ⟶ club fungi (e.g., Agaricus, mushrooms)
- Deuteromycetes ⟶ imperfect fungi (no sexual reproduction observed, e.g., Alternaria)

Phycomycetes – Albugo – Smut = Incorrect (smut fungi belong to Basidiomycetes, not Phycomycetes)

**Final Answer:** (3) Phycomycetes – Albugo – Smut is the incorrect option.

148. **Ans ( 1 )**

NCERT XI Pg # 34, 35, 36, 44

149. **Ans ( 2 )**

**Question Explanation:**

The question asks to identify the substance that appears in the biomacromolecular fraction during cell fractionation but is not truly considered a biomacromolecule.

**Concept:** Biomolecules **Explanation:**

Biomacromolecules are large molecular weight compounds such as:

- Proteins
- Polysaccharides
- Nucleic acids

Although lipids appear in the macromolecular fraction during acid extraction due to their insolubility in water, they are not true biomacromolecules because they do not have polymeric structure like proteins or nucleic acids.

**Final Answer:** (2) Lipid

150. **Ans ( 3 )**

**Question Explanation:**

The question asks to identify the pair of secondary metabolites that are polymeric substances.

**Concept:** Biomolecules – Secondary Metabolites

**Explanation:**

Secondary metabolites are compounds produced by plants, fungi, and microbes that are not directly involved in growth or reproduction.

Some secondary metabolites are polymeric substances, meaning they are made of repeating units:

- Rubber ⟶ polymer of isoprene units
- Gums ⟶ complex polysaccharide polymers

**Final Answer:** (3) Rubber, Gum

151. **Ans ( 4 )**

**Question Explanation:** The question asks to identify the pair of statements that are not correct regarding biomolecules and protein structure.

**Concept:** Biomolecules **Explanation:**

(a) Some secondary metabolites have ecological importance - Correct — many secondary metabolites help in defense and ecological interactions.

(b) NaCl, $CaCO_3$, $PO_4^{3-}$, $CH_3COOH$, CH4 etc are inorganic constituents of living cell - Incorrect — $CH_3COOH$ and $CH_4$ are organic compounds.

(c) Uridine is a nucleoside as well as found in RNA along with phosphate - Correct — uridine is a nucleoside; with phosphate it forms a nucleotide in RNA.

(d) Secondary structure of proteins is absolutely necessary for many biological activities - Incorrect — tertiary structure is mainly responsible for biological activity.

**Final Answer:** (4) d, b

152. **Ans ( 4 )**

**Question Explanation:** The question asks to correctly match different plant growth regulators with their chemical nature or origin.

**Concept:** Plant Growth Regulators – Chemical Nature

**Explanation:**

- Cytokinins ⟶ adenine derivatives
- Ethylene ⟶ gaseous plant hormone
- Gibberellins ⟶ terpene compounds
- Auxins ⟶ indole compounds
- ABA (Abscisic acid) ⟶ carotenoid derivative

**Final Answer:** (4) A-II, B-V, C-IV, D-III, E-I

153. **Ans ( 3 )**

**Question Explanation:** The question asks to identify the enzyme cofactor that is tightly bound to the apoenzyme and may be organic or inorganic in nature.

**Concept:** Biomolecules – Enzymes and Cofactors

**Explanation:**

Enzymes may require non-protein components called cofactors for proper activity.

Types of cofactors:

- Coenzymes ⟶ organic compounds loosely attached
- Metal ions ⟶ inorganic ions assisting enzyme activity
- Prosthetic groups ⟶ tightly bound organic or inorganic components attached to apoenzyme

**Final Answer:** (3) Prosthetic group

154. **Ans ( 4 )**

NCERT-XI, Pg. # 230

155. **Ans ( 4 )**

**Question Explanation:**

The question asks to identify the pair of correct statements related to cyclic photophosphorylation and thylakoid membrane organization.

**Concept:** Photosynthesis – Light Reaction

**Explanation:**

(A) Cyclic photophosphorylation takes place probably to meet the difference in number of ATP and NADPH + $H^+$ used in dark reaction - Correct

(B) Plastoquinone removes many protons from the stroma while transporting an electron - Incorrect — plastoquinone carries specifically 2 protons, so "many protons" is incorrect.

(C) Phosphorylation take place only in light reaction but not in dark reaction - Incorrect — phosphorylation also occurs during Calvin cycle intermediates.

(D) Stroma lamellae have PS-I only and lack NADP reductase - Correct — they mainly possess PSI and are associated with cyclic photophosphorylation.

**Final Answer:** (4) A and D

**156. Ans ( 3 )**

NCERT-XI, Pg. # 85, 91, 90 only c is wrong

**157. Ans ( 2 )**

**Question Explanation:**

The question asks to determine how many of the given tissue structures are absent in a dorsiventral (dicot) leaf.

**Concept:** – Dorsiventral Leaf **Explanation:**

A dorsiventral leaf typically contains:

- Upper and lower epidermis
- Mesophyll (palisade and spongy
- parenchyma)
- Vascular bundles

Structures like hypodermis, endodermis, pericycle, pith, and medullary rays are generally associated with stems or roots, not leaves.

**Final Answer:** (2) Five

**158. Ans ( 1 )**

NCERT Pg. # 70, 77

**159. Ans ( 3 )**

**Question Explanation:**

The question asks to determine the number of progenies producing intermediate sized starch grains in a cross between heterozygous round seeded and wrinkled seeded pea plants.

**Concept:** Principles of Inheritance – Pea Seed Shape

**Explanation:**

- Round seed allele = R (dominant)
- Wrinkled seed allele = r (recessive)

Genotypes and starch grain size:

- RR ⟶ large starch grains
- Rr ⟶ intermediate sized starch grains
- rr ⟶ small starch grains

Cross: Rr * rr Progeny ratio:

- 50% (Rr) ⟶ intermediate starch grains
- 50% (rr) ⟶ small starch grains

Total progenies = 2000

Intermediate starch grain plants: 2000*50100 =1000

**Final Answer:** (3) 1000

**160. Ans ( 4 )**

**Question Explanation:** The question asks to find the probability that the first daughter of a couple will be colour blind.

**Concept:** (Sex-linked Inheritance – Colour Blindness)

**Explanation:**

Colour blindness is an X-linked recessive trait. Let:

- (X) = normal vision allele
- ($X^C$) = colour blind allele

A female becomes colour blind only if she inherits ($X^C$) from both parents.

Both husband and wife have normal vision, but their fathers were colour blind.

Husband's genotype: (X Y)

(A son gets Y chromosome from father, so father's colour blindness does not affect

- him.)

Wife's father was colour blind ($X^CY$), so wife must inherit ($X^C$) from him.

Since she has normal vision, her genotype

- is: $X^CX$

Cross: $X^CX$ * X Y Possible daughters:

- (XX) ⟶ normal
- ($X^CX$) ⟶ carrier, normal

No daughter can be ($X^CX^C$) because father contributes only (X).

Thus, probability of colour blind daughter = 0%

**Final Answer:** (4) 0%

161. **Ans ( 2 )**

**Question Explanation:** The question asks to identify the type of selection involved in developing Indian cattle breeds and the state associated with Sahiwal cows.

**Concept:** Strategies for Enhancement in Food Production – Animal Husbandry

**Explanation:**

Domesticated cattle breeds are developed through artificial selection by humans for desirable traits such as milk yield and disease resistance.

Sahiwal is a famous Indian cattle breed from the Punjab region.

**Final Answer:** (2) (A) Artificial selection ; (B) Punjab

162. **Ans ( 3 )**

NCERT Page - 183

163. **Ans ( 1 )**

**Question Explanation:** The question asks to identify the pests controlled by ladybird and dragonflies.

**Concept:** – Biocontrol Agents

**Explanation:**

Biological control involves using natural predators to control harmful pests.

- Ladybird beetles feed on aphids
- Dragonflies help control mosquitoes

**Final Answer:** (1) a-Aphids, b-Mosquitoes

164. **Ans ( 1 )**

**Question Explanation:** The question asks to determine how many of the given disorders are Mendelian disorders.

**Concept:** Human Genetics – Mendelian Disorders

**Explanation:**

Mendelian disorders are caused by mutation in a single gene and follow Mendelian inheritance patterns.

Examples:

- Haemophilia
- Sickle cell anaemia
- Cystic fibrosis
- Thalassemia
- Phenylketonuria
- Polydactyly

Chromosomal disorders are not Mendelian disorders:

- Klinefelter's syndrome
- Turner's syndrome

Total Mendelian disorders = 6 **Final Answer:** (1) 6

165. **Ans ( 1 )**

**Question Explanation:** The question asks to fill the blanks related to regulation of the lac operon.

**Concept:** Molecular Basis of Inheritance – Lac Operon

**Explanation:**

- In the lac operon, the repressor protein binds to the operator gene and prevents transcription in the absence of lactose.
- This type of control is called negative regulation because the repressor inhibits gene expression.

**Final Answer:** (1) Repressor, Negative

166. **Ans ( 2 )**

**Question Explanation:** The question asks to identify the term used for separation and purification processes carried out after product formation in bioreactors.

**Concept:** Biotechnology – Bioprocess Engineering

**Explanation:**

After a desired product is formed in a bioreactor, it must be:

- Separated
- Purified
- Formulated for marketing

These recovery and purification steps are collectively called downstream processing.

**Final Answer:** (2) Downstream processing

167. **Ans ( 2 )**

**Question Explanation:** The question asks to identify the role of the ampicillin resistance gene used in recombinant DNA technology.

**Concept:** Biotechnology – Tools of Recombinant DNA Technology

**Explanation:**

A selectable marker helps identify and select transformed cells from non-transformed cells.

The ampicillin resistance gene allows only transformed bacteria to survive in the presence of ampicillin.

**Final Answer:** (2) Selectable marker

168. **Ans ( 2 )**

**Question Explanation:**

The question asks to identify the pests controlled by proteins encoded by the cry IAc and cry IIAb genes.

**Concept:** Biotechnology Applications – Bt Cotton

**Explanation:**

Genes from Bacillus thuringiensis produce Bt toxins effective against specific insect pests.

- cry IAc and cry IIAb genes are introduced into Bt cotton
- These toxins are effective against cotton bollworms

**Final Answer:** (2) Cotton bollworms

169. **Ans ( 4 )**

**Question Explanation:** The question asks to identify the correct statement(s) regarding the development of transgenic animals for disease studies.

**Concept:** Biotechnology Applications – Transgenic Animals

**Explanation:**

Transgenic animals are genetically modified animals carrying foreign genes. They are used to:

- Understand the role of genes in diseases
- Serve as models for human diseases
- Test new treatments and therapies

Examples include models for:

- Alzheimer's disease
- Cystic fibrosis
- Cancer and other disorders

**Final Answer:** (4) All of these

170. **Ans ( 1 )**

**Question Explanation:** The question asks to correctly match biotechnology terms with their functions.

**Concept:** Biotechnology – Tools and Techniques

**Explanation:**

- PCR ⟶ used for amplification of DNA/gene
- Bioreactor ⟶ used for large scale culture/production
- Gene gun ⟶ introduces alien DNA into host cells
- Eco RI ⟶ restriction endonuclease enzyme

**Final Answer:** (1) I - D, II - A, III - B, IV - C

171. **Ans ( 1 )**

**Question Explanation:**

The question asks to determine the number of possible genotypes and phenotypes in the ABO blood group system in humans.

**Concept:** Principles of Inheritance – Multiple Alleles

**Explanation:** Possible genotypes:

1. (IA IA)
2. (IA i)
3. (IB IB)
4. (IB i)
5. (IA IB)
6. (ii)

Total genotypes = 6 Phenotypes:

- A
- B
- AB
- O

Total phenotypes = 4 **Final Answer:** (1) 6 & 4

172. **Ans ( 1 )**

**Question Explanation:**

The question asks to correctly match biological control agents with their targets or functions.

**Concept:** Microbes in Human Welfare – Biocontrol Agents

**Explanation:**

- Lady bird feeds on aphids
- Bacillus thuringiensis (Bt) kills butterfly caterpillars
- Baculoviruses infect insects and arthropods
- Trichoderma acts against many plant pathogens

**Final Answer:** (1) a-i, b-ii, c-iii, d-iv

173. **Ans ( 1 )**

NCERT XII Pg # 76, 77

174. **Ans ( 1 )**

NCERT, Page # 53 (E), 53 and 54 (H)

175. **Ans ( 3 )**

**Question Explanation:** The question asks to identify the correct set of statements related to molecular biology and gene expression.

**Concept:** Molecular Basis of Inheritance

**Explanation:**

- RNA is considered the first genetic material in evolution.
- Ribosomes are the site of protein synthesis.
- In lac operon, structural genes are controlled by a common promoter and regulatory system.
- Translation begins when ribosome binds to mRNA, not RNA polymerase.

**Final Answer:** (3) (a), (c) and (d) only

176. **Ans ( 3 )**

**Question Explanation:**

The question asks to correctly match molecular biology terms with their functions/processes.

**Concept:** Molecular Basis of Inheritance

**Explanation:**

- Release factors help terminate translation
- snRNA participates in RNA splicing
- Probe is used in DNA fingerprinting/hybridization
- Rho factor is involved in termination of transcription

**Final Answer:** (3) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

177. **Ans ( 2 )**

**Question Explanation:** The question asks about the function of the z gene of the lac operon.

**Concept:** Molecular Basis of Inheritance – Lac Operon

**Explanation:**

The z gene of the lac operon codes for the enzyme β-galactosidase.

Function of β-galactosidase:

- Hydrolyses lactose (a disaccharide) into:
  - glucose
  - galactose

**Final Answer:** (2) galactose and glucose

178. **Ans ( 4 )**

NCERT XII Page No. # 121, 122

179. **Ans ( 2 )**

**Question Explanation:** The question asks the number of hydrogen bonds formed between complementary nitrogenous bases in double-stranded DNA.

**Concept:** Molecular Basis of Inheritance – DNA Structure

**Explanation:** In double-stranded DNA:

- Adenine (A) pairs with Thymine (T)
- Guanine (G) pairs with Cytosine (C)

Hydrogen bonding:

- A = T ⟶ 2 hydrogen bonds
- G ≡ C ⟶ 3 hydrogen bonds

**Final Answer:** (2) 2, 3

180. **Ans ( 4 )**

NCERT-XII, Pg. # 72, 73